
Where Web Agents Go Blind: Decomposing Reachability from Task Success in Structurally Contained Web Content

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Abstract

1 Web-agent evaluations conflate missing evidence with failures after observation.
2 We introduce a controlled benchmark that separates *reachability* from task success
3 while varying iframe depth and origin, shadow Document Object Model (DOM)
4 containment, and DOM-versus-canvas rendering. Across a 1,760-trial primary grid,
5 every architecture reached every non-canvas target. On canvas, DOM, accessibility-
6 tree, and Chrome DevTools Protocol (CDP) observations fell from 40/40 to 0/40
7 reachability, whereas screenshots remained at 40/40. The original vision deploy-
8 ments completed 0/40 canvas trials: every output contained the target plus an
9 adjacent marker and failed both exact and target-present/marker-absent scoring.
10 Two matched vision-only deployments showed model heterogeneity: gpt-5.4
11 completed 15/20 original-layout canvas trials, while gpt-5-mini completed 0/20.
12 A 240-trial layout follow-up recovered monotonically under both boundary-correct
13 criteria: 0/40 with an inline bar, 0/40 with two spaces, 10/40 with a 40-pixel gap,
14 31/40 on separate lines, and 38/40 at opposite corners. A same-element DOM
15 control also scored 0/40, locating the effect in layout rather than canvas alone.
16 Independent DOM and accessibility probes replicated the iframe null, and blinded
17 recoding reproduced all 40 original failure labels. A final audit found the apparent
18 vision depth effect and variance decomposition confounded by coordinate scaling
19 and absolute position. Observation boundaries are properties of complete, ver-
20 sioned stacks; visible evidence still requires reliable segmentation and coordinate
21 calibration.

22 1 Introduction

23 Web agents now complete multi-step tasks on realistic websites by repeatedly observing a page,
24 selecting an action, and interpreting the resulting state [Deng et al., 2023, Zhou et al., 2024, He
25 et al., 2024]. Observation channels differ: some serialize the Document Object Model (DOM), some
26 consume the accessibility tree, some reason over screenshots, and others traverse browser protocols.
27 Standard execution-based evaluation reports whether the final task succeeded, but it generally assumes
28 that task-relevant evidence was available to the agent. Consequently, the same failure label can mean
29 either that evidence was absent from the observation or that the model misread, mislocalized, or
30 failed to act on available evidence.

31 Modern web structure makes that distinction consequential. Cross-origin iframes separate browsing
32 contexts under the browser same-origin policy; shadow roots encapsulate component subtrees; and
33 canvas exposes pixels without a corresponding textual DOM representation. These mechanisms are
34 common in enterprise and embedded interfaces. They also create security tensions: browser agents

35 with broad cross-origin access can recover useful embedded content, but that access can weaken
36 browser isolation when combined with prompt injection [Roesner and Kohlbrenner, 2026].

37 For agents operating in open-ended environments, such silent observation gaps are both a reliability
38 and an oversight problem: an agent may proceed without knowing that decisive evidence was omitted,
39 while granting broader observation privileges can create new privacy and security exposure. A useful
40 evaluation must therefore distinguish capability under-reach from the separate question of whether an
41 observation channel should be permitted to cross a boundary.

42 Existing benchmarks establish that observation modality matters [Koh et al., 2024], that controlled
43 perturbations reveal hidden robustness gaps [Bai et al., 2026, Yuan et al., 2026], and that canvas can
44 separate visual from structural observations [Levy et al., 2026]. None, however, treats structural
45 reachability as a graded independent variable while holding task content fixed and scoring evidence
46 availability separately from task completion.

47 We make three contributions. First, we introduce a version-pinned structural-containment benchmark
48 whose tasks vary only in iframe depth and origin, shadow-root mode, or rendering target. Second,
49 we report paired reachability and task-success curves, enabling failures before observation to be
50 distinguished from interpretation or execution failures; error coding of archived responses further
51 separates extraction, delimitation, localization, and action-execution failures within the post-capture
52 category. Third, we show through a separation sweep, independent implementation probes, and
53 a runner audit how follow-up controls can distinguish architecture boundaries from layout and
54 coordinate artifacts. The result is deliberately diagnostic rather than leaderboard-oriented: it maps
55 where tested observation pipelines lose evidence and where agents fail despite retaining it.

56 2 Related work

57 WebArena and WebArena Verified provide realistic tasks with execution-based evaluators [Zhou et al.,
58 2024, El Hattami et al., 2025]; VisualWebArena demonstrates that text and multimodal observations
59 yield different outcomes [Koh et al., 2024]. StressWeb and WebForge introduce controlled difficulty
60 and robustness perturbations [Bai et al., 2026, Yuan et al., 2026], while ST-WebAgentBench includes
61 a canvas modality challenge [Levy et al., 2026]. Our benchmark instead holds task semantics fixed,
62 varies structural containment, and scores evidence availability separately from completion.

63 Cross-origin access is also a security choice: consumer agentic browsers implement different policies,
64 and permissive access can enable data theft under prompt injection [Roesner and Kohlbrenner, 2026].
65 We measure capability on self-authored pages, not whether such access should be granted.

66 3 Benchmark and methods

67 3.1 Tasks and structural axes

68 The benchmark contains two deterministic tasks: reading a known value and copying a known
69 value into a target field. Each task first passes a level-0 control consisting of a flat, single-frame,
70 shadow-free, canvas-free page. Generated variants preserve the task instruction, correct answer and
71 marker tokens, and external checker while changing one of four ordered axes:

- 72 1. same-origin iframe depth 0–3;
- 73 2. cross-origin iframe depth 0–3;
- 74 3. shadow containment (none, open, closed, nested open–open); and
- 75 4. rendering target (ordinary DOM or canvas).

76 The shared level-0 condition yields 11 unique structural conditions. Pages and checkers were
77 self-authored and deterministic; no live or authenticated service was included.

78 **Fixed task content.** Across conditions the instruction string, correct answer token, source label, and
79 task-specific marker token are identical. The complete visible strings and layout are not byte- or pixel-
80 identical: DOM pages place the source and prefixed marker in separate elements, whereas canvas
81 draws the source label, value, separator, and unprefixed marker as one line. The experiment therefore

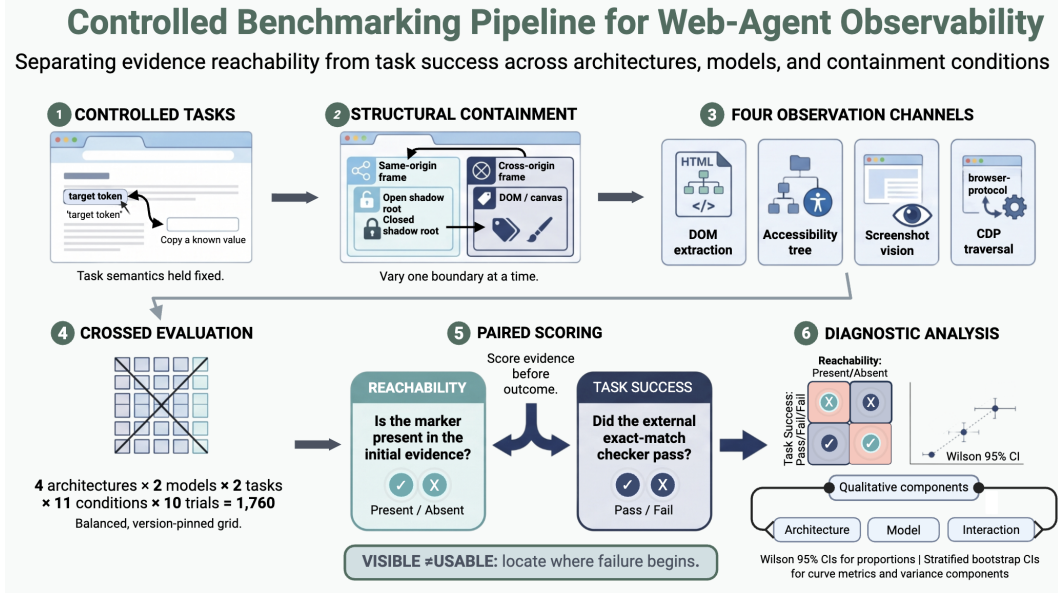


Figure 1: Controlled benchmarking pipeline for web-agent observability. Fixed task semantics are crossed with structural-containment conditions and four observation channels; reachability and exact-match task success are scored separately before diagnostic analysis.

holds task semantics and the two load-bearing tokens fixed while changing both representation and the boundary cues that representation provides.

Fixture geometry. Because the canvas result below turns on visual separation between adjacent strings, we state the fixture layout explicitly. Each task page displays the target value together with a visible task-specific control marker—`CONTROL_READ_VALUE_R2` for reading and `CONTROL_COPY_VALUE_R5` for copying—which is present in every condition and is not part of the correct answer. In DOM conditions, the source label and value occupy one block-level paragraph, while the marker appears in a separate later paragraph in smaller text prefixed by “Target marker:”; the copy form intervenes between those paragraphs in the copy task. In the canvas condition, a single `fillText` call draws the complete string at (20, 80) in a 900×160 canvas using 24-pixel sans-serif type, with the target and marker on the same line separated only by one space, a vertical bar, and one space (|). The copy task requires writing the target value—and only the target value—into a field labeled Destination.

Visual-separation follow-up. To separate rendering modality from layout, a new fixture family holds task, typeface, tokens, and checker fixed while placing the target and marker (i) inline with a bar, (ii) inline with two spaces, (iii) inline with a 40-pixel gap, (iv) on separate lines, or (v) at opposite canvas corners. A sixth control renders the inline-bar string as ordinary DOM text inside one element. Both original vision deployments receive ten trials per task and layout ($N = 240$). The follow-up captures at 1092×768 , matching the high-detail image dimensions delivered to the model so screenshot and action coordinates share one space, and uses a fixed eight-action, 90-second budget. As in the primary sweep, complete timeout transcripts count as task failures; only infrastructure errors are retried and excluded. All 240 allocated outcomes were obtained with no infrastructure exclusions.

3.2 Observation architectures and crossed models

We evaluate four architecture classes defined by their page representation: *DOM extraction* serializes page structure through an automation framework; *accessibility tree* consumes the browser-computed accessibility representation; *vision* receives screenshots and emits coordinate actions; and *Chrome DevTools Protocol (CDP) traversal* consumes browser-level DOM snapshots and frame state. One version-pinned implementation instantiates each class (Appendix A). The labels therefore denote the tested pipelines rather than universal properties of every product using the same modality.

Each architecture is crossed with the same two hosted base models in the primary grid. Every task-model-condition cell contains ten valid trials, giving $4 \times 2 \times 2 \times 11 \times 10 = 1,760$ outcomes. Before the sweeps, a 48-trial control gate required all architecture-model pairs to pass both control tasks three times. In response to the small original roster, a separately versioned vision-only extension adds foundry/gpt-5.4 and foundry/gpt-5-mini on the same two tasks, 11 conditions, and ten-trial allocation (440 additional valid outcomes). Run manifests preserve framework, browser, model deployment, timestamp, task revision, prompt version, and source hashes.

3.3 Outcomes and analysis

Reachability indicates whether the task-specific marker appears in the architecture’s initial task-page evidence; it is rescored from immutable archives without another model call. For the three structure-based architectures this is a text-match test against the archived initial-page serialization. Vision likewise uses no model call: a separate privileged Playwright probe searches every page frame for a benchmark selector (the marker element, target canvas, or shadow host), obtains its bounding box, and scores the trial reachable when at least one nonzero box intersects the 1280×900 captured viewport. For the canvas condition, the operative selector is `#target-canvas`. The probe does not perform optical character recognition or pixel-template matching, and none of its metadata enters the model request. Reachability therefore measures presence of the evidence-bearing fixture region in the observation, and is scored independently of whether the model transcribed, delimited, or acted on it.

Task success is determined by an external state checker using exact match against the correct answer. Strict equality was fixed before the sweeps; Section 4.3 reports a post hoc sensitivity analysis under a relaxed criterion, because the observed canvas failures were compositional rather than substitutional.

For canvas sensitivity analyses we score the emitted answer (read) or submitted field value (copy) under three content criteria: exact match after outer-whitespace stripping; target present as a substring; and target present with the task-specific marker absent. The third permits explanatory wrapper text but rejects the observed target-marker concatenation.

We additionally record actions, input tokens, output tokens, elapsed time, and four joint outcomes formed by crossing reachability with success. Success conditional on unreachable evidence is reported only as an operational confabulation proxy, not as a claim about model intent.

Failure coding. For every failed vision trial we additionally code, from the archived model response and action log, where in the pipeline the failure occurred: extraction (the correct answer token is absent or altered in the response), delimitation (the correct token is present but bounded incorrectly, so the emitted string is a superset or subset of the answer), localization (the response is correct but coordinate actions target the wrong element), action execution (correct content and target, but the emitted action sequence does not reach the checked state), and abstention. Coding operates on preserved archives and requires no additional model calls.

Proportions use two-sided Wilson 95% confidence intervals (CIs). Reachability estimates and Figure 2A pool the two tasks and two primary models within an architecture ($N = 40$ per condition), as reachability is scored without a model call. Figure 2B combines the primary and extension deployments on the matched original-layout vision-canvas cells while pooling tasks ($N = 20$ per deployment); Appendix B reports the underlying task-model cells ($N = 10$). The four-model vision results are otherwise kept deployment-resolved, and the extension models do not enter architecture comparisons because they were not run on the three structure-based stacks. Curves use normalized trapezoidal area under the curve (AUC), total level-0-to-final drop, largest adjacent drop, and a preregistered cliff indicator. Curve-shape variance is decomposed with a balanced two-way sum of squares over architecture, model, and their interaction; 2,000 stratified bootstrap resamples produce CIs. With only two models, this decomposition is descriptive of the tested roster rather than a population random-effects estimate. Full definitions appear in Appendix C.

4 Results

All four sweeps were complete and balanced at 440 valid outcomes per architecture. The control gate passed 48/48 trials. A provider-free rescore of the DOM and accessibility-tree archives under the final initial-page policy changed no trial labels.

Table 1: Overall outcomes and recorded resource use across all 11 conditions ($N = 440$ per architecture). Tokens are mean input plus output tokens; provider-dollar cost was not present in archived responses. Success is scored by exact match; see Section 4.3 for a relaxed-criterion sensitivity analysis of the vision canvas cells.

Architecture	Reachability	Success	Actions/run	Tokens/run
DOM extraction	400/440	400/440	2.46	14,839
Accessibility tree	400/440	398/440	2.03	14,927
Vision	440/440	357/440	4.16	8,645
CDP traversal	400/440	399/440	2.71	8,765

4.1 Reachability was flat except at the rendering boundary

Every architecture exposed the target marker in 40/40 trials for every non-canvas condition (Wilson 95% CI 91.2–100%). Thus neither iframe origin or depth nor open, closed, or nested shadow containment caused measured reachability loss in the tested implementations. Because structure-based reachability is scored by text match and vision reachability by a privileged Playwright probe, neither score requires or depends on a model call. Pooling reachability over models is therefore appropriate by construction. A targeted replication then tested the iframe cells using Selenium recursive frame switching as a second DOM implementation and Playwright’s public per-frame Accessible Rich Internet Applications (ARIA) snapshot as a second accessibility implementation. Each reached 16/16 task–origin–depth cells. The null therefore held in both tested DOM implementations and both tested accessibility implementations, rather than only in browser-use and one raw-CDP accessibility collector. This result contradicts the cross-origin and closed-shadow cliffs predicted by the mechanistic account that motivated the design.

Canvas produced the sole reachability separation (Figure 2A). DOM extraction, accessibility tree, and CDP traversal each fell from 40/40 on the ordinary DOM target to 0/40 on canvas (0%; 95% CI 0–8.8%). Vision remained at 40/40 because the target pixels remained in the screenshot. Under the cliff rule, every task–model curve in the three structure-based architectures contained a complete canvas cliff; vision contained none.

4.2 Task success separated visibility from execution

Success closely tracked reachability for structure-based observations. Excluding canvas, DOM extraction completed 400/400 trials, accessibility tree 398/400 (99.5%), and CDP traversal 399/400 (99.8%); all three then completed 0/40 canvas trials. Among the 40 unreachable trials for each of these architectures, none succeeded, giving a confabulation-proxy estimate of 0% (95% CI 0–8.8%). Vision had no unreachable trials, so its conditional proxy is undefined rather than zero.

Vision exposed the complementary failure mode (Figure 2B). It reached the canvas target in all 40 trials but completed none (0%; 95% CI 0–8.8%), and it completed all 120 shadow-containment trials. The original iframe completion rates varied non-monotonically despite 100% reachability, but inspection of all 40 anomalous gpt-5.6-terra screenshots and action logs shows that this pattern is not evidence of a depth mechanism. The application programming interface’s (API’s) high-detail preprocessing scaled the 1280×900 capture to approximately 1092×768 , while returned coordinates were executed in the original viewport. At depth 1, all 20 first copy clicks landed at $y = 224$ – 226 , in the 12-pixel gap between the destination label and field. At depth 2, the same biased coordinate happened to land inside the associated label and focused the input, producing an accidental recovery. The depth-3 read target was fully visible, but each of the two condition-specific screenshots elicited the identical mis-transcription AI PHA-7391 in all ten repeats. Repeats therefore sampled model behavior on identical pixels and fixed absolute positions. Appendix B retains the observed cells for transparency, but we treat them as a coordinate-runner audit and make no depth claim from them.

The two-deployment vision extension completed all 440 allocated trials with 440/440 reachable targets. Combined with the matched primary vision cells in Figure 2B, it shows that the original canvas failure is not universal across models: gpt-5.4 completed 15/20 original-layout canvas trials, whereas gpt-5-mini completed 0/20, matching the two original deployments’ 0/20 each. The pooled four-model canvas rate is therefore 15/80, but the deployment-resolved result is primary because pooling masks this heterogeneity. Across all 11 conditions, gpt-5.4 completed 169/220 trials and

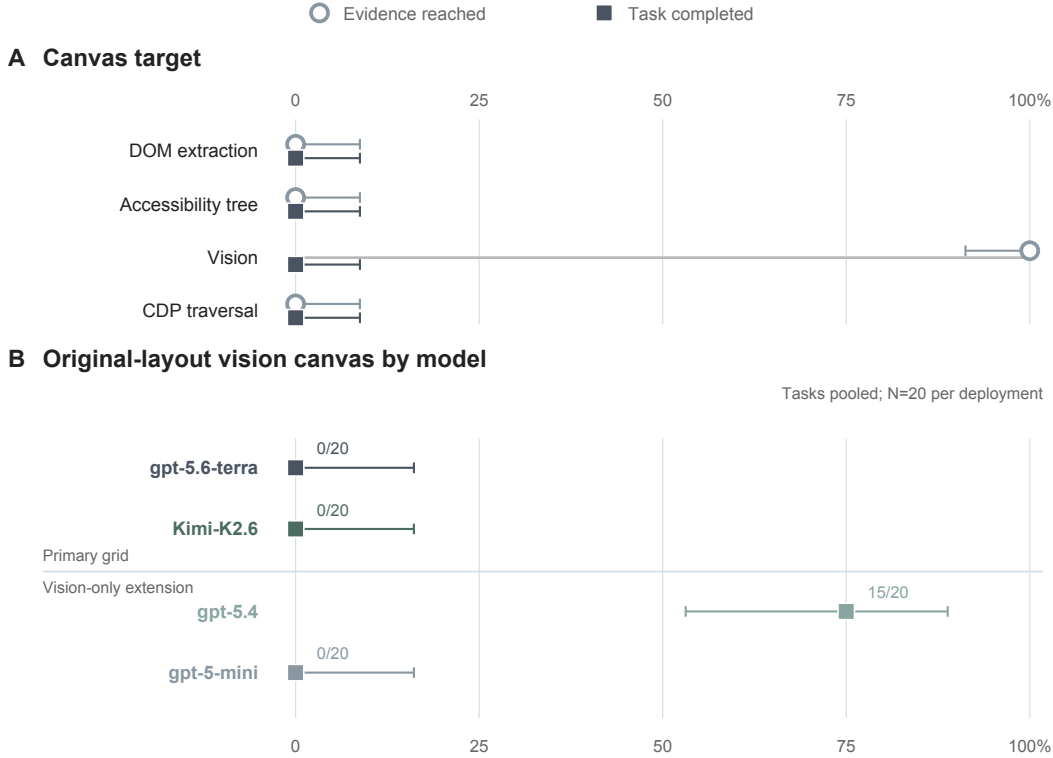


Figure 2: **Evidence availability and task completion separate under visual observation.** (A) Primary-grid canvas estimates pool two tasks and two models ($N = 40$). Structure-based observations neither reached nor completed the task; vision reached 40/40 targets but completed 0/40 under exact match. (B) The matched original-layout vision-canvas cells combine the two primary and two extension deployments while retaining deployment-level estimates ($N = 20$ each): gpt-5.4 completed 15/20, whereas the other three completed 0/20. Whiskers are Wilson 95% confidence intervals; Appendix B gives the $N = 10$ task-model cells and the full four-deployment vision grid.

gpt-5-mini 189/220. Their model-task-condition cells (Appendix B) remain position-sensitive: for example, gpt-5.4 copy success was 2/10, 0/10, and 7/10 at same-origin depths 1–3 while its read success was 10/10 throughout. The broader roster demonstrates model heterogeneity but does not rehabilitate a depth interpretation of fixed-position fixtures.

4.3 Canvas failure was a delimitation failure, not blindness

Coding the 40 failed canvas trials from the two original deployments against the archived model responses and action logs, both deployments failed all 20 trials each. Across all 40, 40/40 misreported the exact target value, 0/40 reported it correctly but mislocalized the `Destination` field, 0/40 reached correct extraction and localization but failed during action execution, and 0/40 refused or abstained. Every response appended the visible control marker to the true token; in all 20 copy trials, the checker recorded that composite string in the `Destination` field. Within that preregistered roster, failure was entirely extraction—specifically, target delimitation—rather than grounding or action execution. The later gpt-5.4 successes show that the behavior is not model-universal.

The original failure was compositional rather than substitutional: the correct token was recovered from pixels in every coded trial and was then over-included rather than replaced or hallucinated. The follow-up below identifies the operative factor more narrowly: close visual layout makes the answer boundary ambiguous, while structural serialization can supply a boundary when the tokens occupy distinct elements. Canvas is one way to remove those structural cues, not a sufficient cause of failure by itself.

Visual separation produces a dose-response in delimitation

Pooled over two tasks and two deployments; Wilson 95% intervals

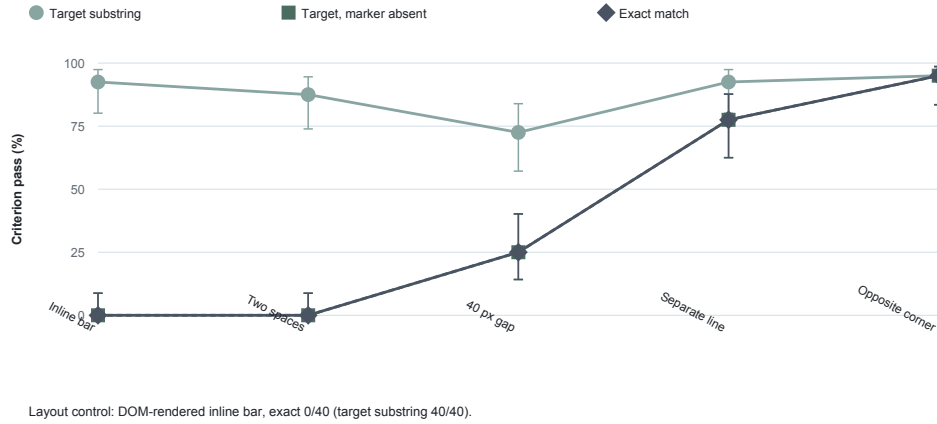


Figure 3: **Boundary-correct output rises with visual separation.** Each point pools two tasks, two deployments, and ten trials per task–deployment cell ($N = 40$); whiskers are Wilson 95% intervals. Exact match and the target-present/marker-absent criterion coincide. The separate DOM inline-bar control is reported in the footer because it is not a sixth separation level.

Checker sensitivity. Because all 40 responses contained the target token as a substring, the canvas success rate depends jointly on rendering and on scoring strictness. We therefore report three criteria: the preregistered exact match (0/40), a permissive substring check (40/40), and an intermediate check allowing wrapper text but requiring that the target appear and the task-specific marker be absent (0/40). The last criterion rejects precisely the observed boundary error without requiring bare-token output. Thus the claim that these trials failed to delimit target from marker does not depend on choosing exact match over substring; substring instead serves as evidence that the target glyphs were read.

Separation produces a dose-response; rendering alone does not. Figure 3 reports the controlled follow-up. Exact-match and target-without-marker criteria agreed at every level and increased monotonically from 0/40 for the inline bar, to 0/40 for two spaces, 10/40 for a 40-pixel gap, 31/40 for separate lines, and 38/40 for opposite corners. The corresponding Wilson 95% intervals were 0–8.8%, 0–8.8%, 14.2–40.2%, 62.5–87.7%, and 83.5–98.6%. This is the requested dose-response: increasing visual separation recovers boundary-correct output. The same-element DOM inline-bar control also scored 0/40 under exact and target-without-marker criteria while retaining the target substring in 40/40. Thus the original failure is not uniquely caused by canvas rendering; it is a layout-driven delimitation failure that canvas serialization cannot repair with element boundaries. Under the permissive substring criterion, the five canvas levels scored 37/40, 35/40, 29/40, 37/40, and 38/40; the nonmonotonic values reflect complete timeout/failure transcripts concentrated in Kimi copy trials rather than target–marker boundary acceptance.

4.4 The original variance estimate is runner-confounded

The registered analysis assigned 100% of observed reachability curve-shape variance to architecture and partitioned task-success shape as architecture 48.7% (95% CI 43.0–54.9%), model 13.9% (11.2–17.0%), and architecture–model interaction 37.4% (31.7–43.1%). Those arithmetic summaries are reproducible, but the screenshot/action audit changes their interpretation. Structure-based agents were at ceiling outside canvas, so nearly all non-deterministic success variance came from the vision cells; those cells in turn contain the coordinate-scale and absolute-position confounds described above. The bootstrap cannot remove a deterministic fixture bias by resampling identical screenshots. We therefore retain the decomposition only as a descriptive audit of the original versioned stacks and do not interpret its percentages as architecture, model-family, or depth effects. The two additional vision deployments broaden the observed heterogeneity, including a 15/20 versus 0/20 canvas contrast, but a

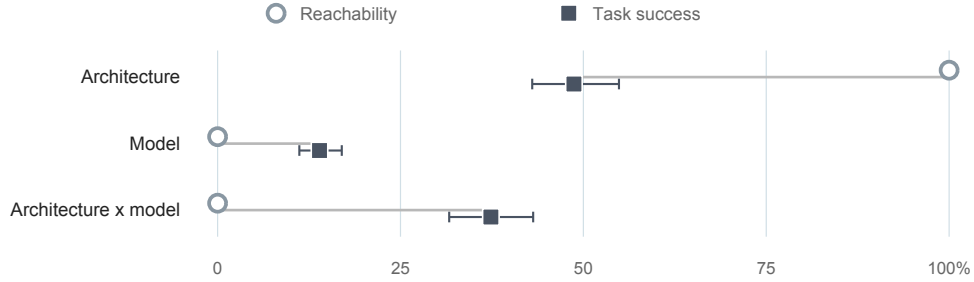


Figure 4: **Registered decomposition of the original grid, retained as a runner audit.** Points show each component’s share of the balanced two-way sum of squares; horizontal whiskers are stratified-bootstrap 95% intervals. Post hoc inspection identified coordinate-scale and absolute-position confounds in the vision cells that supply nearly all non-deterministic variation, so these shares are not interpreted as causal architecture, model-family, or depth effects.

vision-only extension cannot form a balanced architecture-by-model decomposition. A valid causal variance comparison requires calibrated coordinates, position-matched or randomized fixtures, and the broader roster crossed with every architecture under analysis.

4.5 Observation channels had different resource profiles

Resource use did not follow a simple “vision is more expensive” pattern. Vision took the most actions but used fewer tokens overall and per action (2,076 tokens/action) than DOM extraction (6,034), accessibility tree (7,338), or CDP traversal (3,230). Per-action figures are means of per-trial ratios and therefore differ slightly from the ratio of the pooled means in Table 1. Accessibility-tree tokens per action rose most sharply with iframe depth, consistent with increasingly large serialized tree contexts. Appendix E shows the pooled iframe-depth curve; the artifact includes the complete per-condition cost curves. Because archived responses omitted billing amounts, we report actions and tokens rather than imputing dollar cost.

5 Discussion

The follow-ups change the interpretation more than the headline direction. Canvas does not universally defeat vision: rasterization removes explicit element boundaries and makes models reconstruct them from layout, with success varying by separation and deployment. Likewise, the iframe result is an implementation-level null across the tested collectors, not evidence that every DOM- or accessibility-based agent crosses every browser boundary.

The tested iframe and shadow conditions removed no evidence, while canvas removed textual evidence from all three structure-based stacks but not from screenshots. The iframe null contradicts the mechanistic expectation that same-origin restrictions on page scripts automatically constrain agent observations. Automation frameworks and browser-protocol layers can expose content across those boundaries, so architecture labels must denote versioned implementations rather than universal modality properties. This result does not erase browser security boundaries: it separates the capability measured by an unrestricted collector from the normative question of whether that access should be granted.

For vision, availability was not usability. The original deployments read the target glyphs but appended the marker; increasing separation repaired the error, and a same-element DOM control reproduced it. The mechanism is visual delimitation under this layout, not an intrinsic inability to read canvas. This matters operationally: a model that has recovered the right symbols but bounded them incorrectly needs boundary clarification, not a different observation channel. Region prompts, structural cross-checks, or answer-format constraints are therefore more targeted remedies than abandoning screenshots, and the gpt-5.4 result shows that the boundary can also be resolved by a stronger deployment. The absence of success when evidence was unavailable further supports using unpredictable target tokens to detect unsupported completion.

290 The coordinate audit prevents a causal reading of the registered variance shares. Bootstrap inter-
291 vals quantify stability under resampling of the archived fixtures, but they cannot remove a fixed
292 image-scaling error or absolute-position bias; precision is not construct validity. A credible vari-
293 ance comparison would require calibrated coordinates, randomized or position-matched fixtures,
294 and the broader model roster crossed with every architecture. Likewise, the cost results show no
295 uniformly best channel—vision used more actions but fewer tokens, while serialized observations
296 exposed different blind spots. This motivates selective hybrid observation rather than a single default
297 representation.

298 These results also argue against ranking channels by one aggregate score. Reachability diagnoses
299 capture; the three response criteria expose boundary–grading interactions; actions and tokens measure
300 different efficiencies. Collapsing them makes unavailable, misdelimited, and execution-failed targets
301 equivalent. Reports should retain this decomposition and attach claims to identifying interventions:
302 containment for reachability, separation and matched DOM layout for delimitation, implementation
303 replication for generality, and calibrated randomized fixtures for model comparisons.

304 **6 Limitations**

305 Two synthetic tasks isolate mechanisms but do not establish performance on long-horizon websites
306 or combined structural barriers. Each architecture remains one versioned stack, although independent
307 DOM and accessibility probes replicated the iframe null. The two added deployments cover only
308 vision, so they reveal model heterogeneity without supporting a four-model architecture comparison;
309 the coordinate confound independently makes the registered variance shares non-causal. Marker
310 presence establishes availability, not successful interpretation or action.

311 The grid omits permission-respecting real-page and restricted-security arms, provider-dollar cost,
312 and machine-level energy. The original canvas fixture also coupled rendering with close layout; the
313 separation sweep, DOM control, and three scoring criteria address but do not exhaust that design
314 space or establish a universal separation threshold. The zero-success confabulation proxy has an
315 8.8% upper Wilson bound, so it is evidence against frequent unsupported success rather than proof
316 of impossibility. Failure labels infer stages from archives rather than latent states; the blinded coder
317 agreed on 40/40 cases, but ordinary Cohen’s κ is undefined because both coders used one category
318 (prevalence-adjusted bias-adjusted $\kappa = 1.00$ only as a sensitivity statistic).

319 **7 Ethics and broader impacts**

320 All URLs and assets were self-authored synthetic fixtures. The motivating healthcare portal was never
321 accessed or included; no personal data, third-party accounts, human participants, or crowdsourced
322 labor were used. Broader cross-origin reach may enable data exfiltration under prompt injection, so
323 capability should be separated from permission and paired with access controls, provenance logging,
324 and evaluation of restricted versus unrestricted modes before deployment.

325 **8 Conclusion and future work**

326 Across the primary grid, iframe and shadow containment caused no reachability loss, while canvas
327 separated screenshot from structural evidence. The follow-ups locate the original vision failure in
328 layout-sensitive delimitation, show model heterogeneity, and invalidate causal depth or variance
329 claims from the uncalibrated runner. Evaluations should therefore report reachability, completion,
330 provenance, and cost separately and treat observation boundaries as properties of complete versioned
331 stacks.

332 Future work should test permission-respecting real or sandboxed sites and combine structural barriers
333 rather than varying them one at a time. Capability under-reach should be evaluated jointly with
334 security over-reach using adversarial instructions and sensitive decoy content. Hybrid agents should
335 be tested on whether they detect uncertainty and invoke region prompts or structural cross-checks
336 only when needed, recovering evidence without paying the token and latency costs of every channel.
337 Finally, a calibrated-coordinate, position-randomized vision rerun is required before re-estimating
338 model or depth variance.

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368 Supplementary Material

369 A Implementation and run provenance

Table 2: Version-pinned implementations used in the completed grid. Model deployment identifiers are the aliases recorded in the manifests.

Class	Implementation	Prompt/version
DOM extraction	browser-use 0.13.7 at commit f0aa3a8; Chromium 1228	dom-no-vision-v2
Accessibility tree	Playwright 1.61.0; Chromium 1228; CDP accessibility snapshot	ax-full-action-v3
Vision	Playwright 1.61.0 screenshot and coordinate-action runner; Chromium 1228	vision-coordinate-v1
CDP traversal	Playwright 1.61.0 CDP DOM-snapshot runner; Chromium 1228	cdp-snapshot-v1

370 The two primary Azure AI Foundry deployments were recorded as foundry/gpt-5.6-terra
 371 and foundry/Kimi-K2.6. The vision-only extension added foundry/gpt-5.4 and
 372 foundry/gpt-5-mini; its 440/440 valid outcomes are hash-bound to the primary vision resolved
 373 specification. Primary-grid execution ran from 2026-08-30 17:55:16 to 2026-08-31 02:17:28 UTC,
 374 and the provider-free reconciliation manifest was written at 02:33:59 UTC. The 2026-09-04 follow-
 375 ups have separate immutable windows: the implementation probes ran 14:38:00–14:38:04 UTC, the
 376 240-trial separation sweep 14:51:51–15:10:19 UTC, and the 440-trial vision extension 15:28:33–
 377 16:02:50 UTC. These dates are reported separately rather than retroactively changing the primary
 378 receipts. The DOM, accessibility, vision, and CDP receipts each contain task revisions, condition and
 379 model rosters, valid-trial counts, source and result hashes, exclusions, and completion verdicts. The
 380 final primary control gate used the same four architecture classes and two models and passed all 48
 381 required trials.

382 For the iframe replication, Selenium 4.48.0 recursively switched into each frame and inspected that
 383 frame’s own page_source; Playwright 1.61.0 independently obtained a public aria_snapshot
 384 from each frame. Neither path reused browser-use’s serialization or the primary accessibility runner’s
 385 raw Accessibility.getFullAXTree call. Both reached all 16 task–origin–depth cells.

386 B Model-resolved task-success cells

387 The headline primary-grid results remain pooled where the split is uninformative. Table 3 confirms
 388 that both primary deployments were at or near ceiling for every non-canvas structure-based archi-
 389 tecture. Figure 2B combines all four deployments on the matched original-layout vision-canvas
 390 cells; Table 4 reports the complete four-deployment vision grid. The extension models were not run
 391 through the three structure-based agent stacks and therefore do not enter an architecture-by-model
 392 decomposition.

Table 3: Non-canvas task success for the three structure-based architectures, shown by model deployment. Each model column contains $N = 200$ trials; the pooled column contains $N = 400$.

Architecture	gpt-5.6-terra	Kimi-K2.6	Pooled
DOM extraction	200/200	200/200	400/400
Accessibility tree	200/200	198/200	398/400
CDP traversal	200/200	199/200	399/400

393 C Statistical definitions

394 For a curve with ordered rates $y_0, \dots, y_L$, normalized AUC is the equally spaced trapezoidal area
 395 divided by L ; total drop is $y_0 - y_L$; and the largest single-level drop is $\max_\ell(y_{\ell-1} - y_\ell)$. The cliff
 396 ratio is the largest adjacent drop divided by total drop. Curves with total drop below 0.20 are labeled

Table 4: Complete four-deployment vision iframe task-success grid. Entries are successes out of ten trials. S denotes same-origin and X cross-origin; depth 0 is the shared level-0 control and is repeated across the two blocks for comparison. The bottom four rows are the vision-only extension.

Model deployment	Task	S0	S1	S2	S3	X0	X1	X2	X3
foundry/gpt-5.6-terra	Copy	10/10	0/10	10/10	10/10	10/10	0/10	10/10	9/10
foundry/gpt-5.6-terra	Read	10/10	10/10	10/10	0/10	10/10	10/10	10/10	0/10
foundry/Kimi-K2.6	Copy	10/10	10/10	10/10	10/10	10/10	10/10	10/10	8/10
foundry/Kimi-K2.6	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
foundry/gpt-5.4	Copy	10/10	2/10	0/10	7/10	10/10	0/10	7/10	2/10
foundry/gpt-5.4	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
foundry/gpt-5-mini	Copy	10/10	10/10	9/10	7/10	10/10	9/10	9/10	8/10
foundry/gpt-5-mini	Read	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10

no_material_net_drop; remaining curves are labeled cliff when the ratio is at least 2/3 and slope otherwise. Threshold sensitivity at 0.50 and 0.75 was also computed.

Wilson intervals use the two-sided 95% score interval. Curve-metric and variance-component intervals use a deterministic stratified nonparametric bootstrap: trials are resampled with replacement within each task-model-condition cell, and the complete statistic is recomputed for 2,000 resamples. Where a contrast is deterministic within cells, as for the canvas reachability split, the resulting interval is degenerate and carries no information about sampling variability. Curve shape for the variance decomposition is represented by level-0-minus-level rates at every nonbaseline axis level, averaged over the two tasks. A balanced two-way sum of squares partitions variation into architecture, model, and interaction shares, normalized by their sum.

D Failure coding scheme

Each failed vision trial is assigned exactly one category, applied in order, from the archived response text and action log:

1. **Extraction—substitution:** the correct answer token does not appear in the response.
2. **Extraction—delimitation:** the correct answer token appears, but the emitted answer is a strict superset or subset of it (for example, an adjacent visible string is appended).
3. **Localization:** the emitted answer is exactly correct, but coordinate actions targeted an element other than the specified field.
4. **Action execution:** content and target are both correct, but the emitted action sequence did not produce the checked state.
5. **Abstention:** the model declined, reported that it could not find the target, or emitted no answer.

Categories are mutually exclusive and jointly exhaustive over failed trials. Coding operates only on preserved archives and involves no additional provider calls, so it inherits the provider-free reproducibility of the rest of the analysis. Initial labels were assigned deterministically by exact-string and substring comparison against the known target and marker, then hand-checked against all 40 archived response and action records. A second coding agent independently received a randomized packet with opaque case identifiers and was blinded to model, condition, repeat, run identifier, first-coder labels, and aggregate results. It assigned all 40 cases to extraction failure, matching the first coder in 40/40 cases. Ordinary Cohen’s κ is undefined because both marginal label distributions have zero variance (expected chance agreement is 1.0); raw agreement is 100%, and prevalence-adjusted bias-adjusted $\kappa = 1.00$ is reported only as a sensitivity statistic.

429 E Recorded resource curves

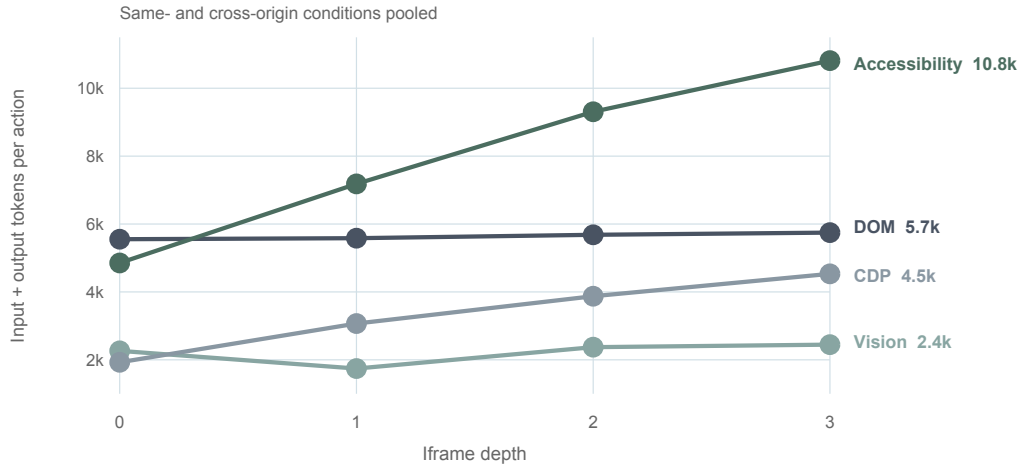


Figure 5: **Serialized context grows differently across observation channels.** Lines show mean input-plus-output tokens per recorded action across iframe depth, pooling same- and cross-origin conditions, tasks, and models. The shared depth-0 control has $N = 40$ per architecture; depths 1–3 have $N = 80$ each. Overall actions and tokens per run are reported in Table 1; complete per-condition estimates are included in the artifact.

430 F Reproducibility notes

431 Code, benchmark fixtures, archived outcomes, and analysis artifacts are available in an anonymous
 432 repository: <https://anonymous.4open.science/r/web-agents-cant-see-61DC/>.

433 The repository contains the self-authored fixtures, task specifications, frozen model roster, four
 434 runners, immutable per-trial archives, sweep receipts, and analysis outputs. A complete reproduction
 435 requires a supported Chromium installation, Python dependencies pinned by the repository lockfile,
 436 and access to the two recorded hosted model deployments. The core analysis is provider-free once
 437 archives exist: reachability rescoring, Wilson intervals, cliff classification, failure coding, variance
 438 decomposition, and figure generation operate only on preserved artifacts. API credentials are supplied
 439 through environment variables and are excluded from manifests and source control.

440 Experiments ran from a local Apple M2 machine with eight processor cores and 16 gigabytes of
 441 unified memory; model inference was provided by hosted Azure AI Foundry deployments, whose
 442 server hardware was not exposed. Across the 1,760 primary-grid trials, archived elapsed times sum
 443 to 34,253 seconds (9.51 trial-hours), with a mean of 19.46 seconds per trial. Architecture means were
 444 20.00 seconds for DOM extraction, 19.14 for accessibility tree, 18.27 for vision, and 20.45 for CDP
 445 traversal.

446 **Large language model (LLM)-assisted development.** An LLM-based coding agent was used
 447 to assist with implementation and refactoring of benchmark runners, scoring and analysis utilities,
 448 figure generation, and repository packaging. The authors specified the study design, executed the
 449 experiments, reviewed the generated artifacts, and retain responsibility for the correctness of the code,
 450 results, citations, figures, and claims.